

Numerical Analysis – Homework #2

Utah State University Fall 2026

1.

Perform the following calculations using 3-digit rounding, 3-digit chopping and exactly:

(a) $12.3 + 0.0234$

(b) $-0.0321 + 0.000136$

(c) $12.3 - 0.0234$

2.

Write the polynomial

$$p(x) = 5x^6 + x^5 + 3x^4 + 3x^3 - x^2 + 1$$

in nested form.

3.

Suppose p^* must approximate p with relative error at most 10^{-3} . Find the largest interval in which p^* must lie for each value of p .

(a) $p = 150$

(b) $p = 1500$

4.

Use 3-digit chopping (after each addition) for

$$\sum_{i=1}^N \frac{1}{i}.$$

(a) Find N such that N is the smallest integer for which your sum is not the exact solution (chopped to 3 digits).(b) What is the absolute error for this value of N ?(c) What is the relative error for this value of N ?

5.

The number e can be calculated as

$$e = \sum_{n=0}^{\infty} \frac{1}{n!}.$$

Use three-digit chopping arithmetic to compute the following approximations to e . Also compute the relative and absolute errors. For the 'exact' value of e , use $e = 2.7182818$.

(a) $S_1 = \sum_{n=0}^5 \frac{1}{n!}$

(b) $S_2 = \sum_{n=0}^5 \frac{1}{(5-n)!}$